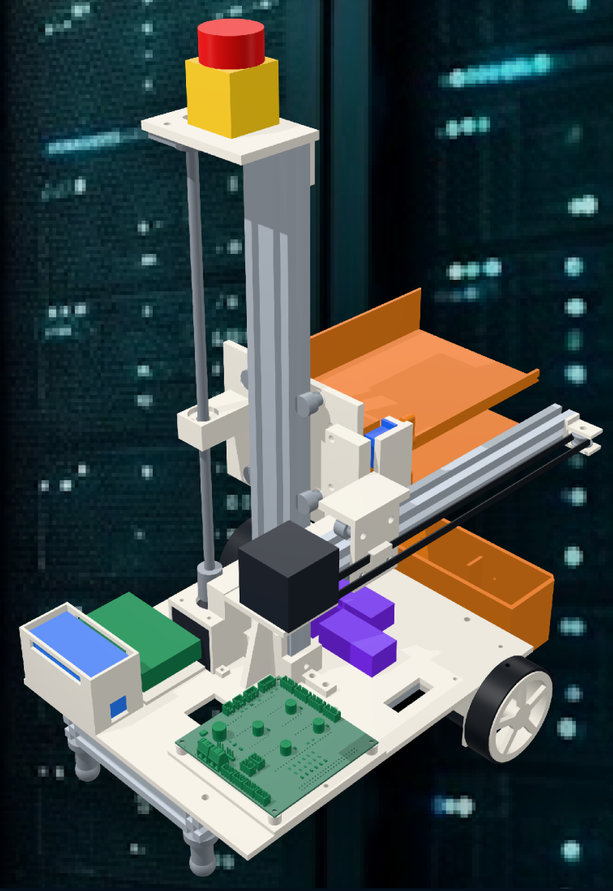
AUTOMATION TRACK

DELTA CUP 2026

CORVIN

Reducing manual downtime in data centers with an autonomous robot that replaces failed hard drives, with real-time operator control



THE PROBLEM

Drives fail, people walk

Until a technician swaps the drive, the RAID array runs without redundancy.

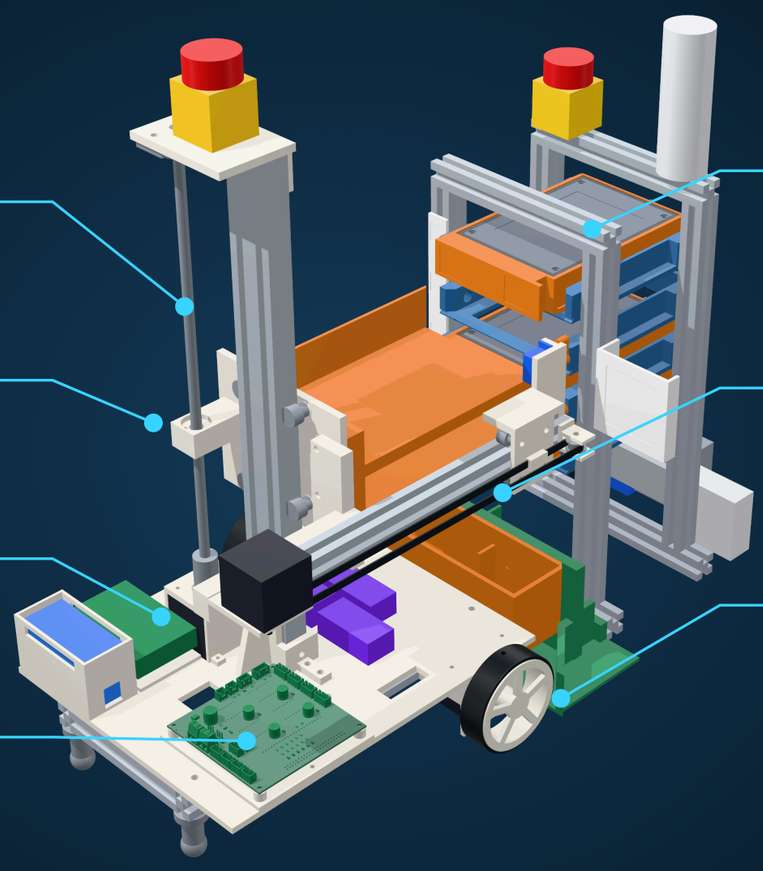
OUR SOLUTION

Corvin swaps them itself

It fetches a spare, swaps the drive and returns the failed one for RMA.

Corvin at the rack

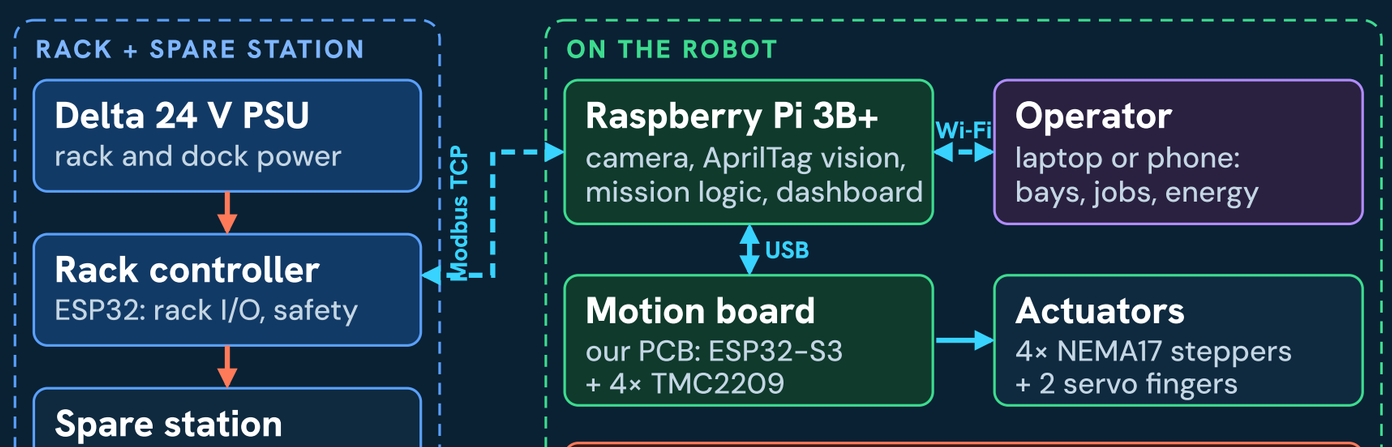
CAD model · docked at a 4-bay rack



One swap, start to finish

1. **Detect** — Dashboard flags a failing drive in bay N.
2. **Pick the spare** — Pulls a good drive into the upper slot.
3. **Drive and dock** — AprilTag approach, V-funnel to ±2 mm.
4. **Extract** — Checks the caddy tag, pulls the failed drive.
5. **Insert** — Lift drops 45 mm, pushes the spare in.
6. **Return for RMA** — Leaves the failed drive in the RMA slot.
7. **Charge and log** — Charges on the dock, logs time and Wh.

System architecture



DELTA INSIDE

Industrial-style control

- Delta 24 V PSU: rack and dock power
- ESP32 controller: 24 V I/O, safety
- Same Modbus TCP map as a Delta PLC

IIOT + ENERGY

Every swap, live

- Bays, jobs and camera, 5 updates/s
- Energy (Wh) measured per swap
- Sleeps on the dock between jobs

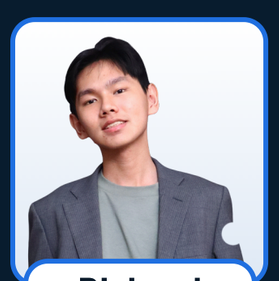
≤3 min per full swap (107 s simulated)

48/50 swaps with no help: our test target

±2 mm docking repeatability with the V-funnel

2 trips per job: home → rack → home

50 Wh battery, under the 100 Wh flight limit


Richard


Saleh


Bagus


Rangga

TEAM

Boys Go To Thailand

University of Brawijaya, Indonesia